



CAEAP Response for Penn State EA Advisory Board

What **competencies** should an undergraduate graduating with a bachelors in enterprise architecture possess? To what degree or level should this student be proficient in each competency area? What mechanisms can we use to assess this degree of proficiency?

Competency Matrix

Focus Area	Competency	Proficiency	Measure
Leadership	Create and Sustain Mission and Strategy	Knowledgeable	
	Estimating and Managing for Profit	Competent	
	Managing and Developing Talent	Knowledgeable	
	Building High Performance Commitment	Competent	
Business	Leveraging Industry and Market Knowledge	Competent	
	Customer/Client Knowledge	Knowledgeable	
	Company and Product Knowledge	Knowledgeable	
	Communication/Interpersonal Skills	Competent	
	Planning/Project Management Skills	Competent	
	Analytical Problem-Solving and Decision Making Skills	Competent	
	Data Interpretation and Analysis Skills	Competent	
	Concern for Detail and Quality	Competent	
	Consultative Selling Skills	Knowledgeable	
	Marketing Skills	Knowledgeable	
	Functional/Technical Knowledge	Competent	
	Develop Strategy	Competent	
Architectural	Create Influence	Competent	
	Develop Enterprise Models	Competent	
	Identify Opportunities for Alignment	Competent	
	Develop Metrics Models	Competent	
	Define & Build Governance Frameworks	Competent	
	Define Fiduciary Responsibilities	Competent	
General	Continuous Learning	Competent	
	Collaboration / Teamwork	Competent	
	Innovation & Continuous Improvement	Competent	
	Focus on Customer Value	Competent	



What are the top 10 areas of specialization that we should offer (i.e. security, information management, networking, supply chain, etc.)? How should these be prioritized if we have resource restrictions that allow us to offer only some (say, 2 or 3) of these? This specialization would take the form of 2 or 3 course curricular tracks.

General & Specialization Matrix

Focus Area	Skill Areas	Prioritization
Application Architecture	Build vs. Buy Analysis	H
Application Architecture	Business Services (Process Automation)	H
Application Architecture	Component Design & Engineering	M
Application Architecture	Designing for Architecture "ilities"	H
Application Architecture	Solution Engineering (Sourcing, Open Source, SaaS, Software Engineering)	H
Business Architecture	Business Case Development	H
Business Architecture	Business Metrics Modeling	H
Business Architecture	business modeling	H
Business Architecture	Business Performance Modeling	H
Business Architecture	Business Process Modeling	H
Business Architecture	Business Productivity Optimization	M
Business Architecture	Business Roadmapping	H
Business Architecture	capital planning	L
Business Architecture	Creating Vision	M
Business Architecture	Customer Focus Methodology	H
Business Architecture	Defining Principles	M
Business Architecture	Developing Strategy	H
Business Architecture	Driving Innovation	L
Business Architecture	Enterprise alignment	H
Business Architecture	Enterprise Governance Management	M
Business Architecture	Identifying desired outcomes	H
Business Architecture	Influencing business planning	H
Business Architecture	Marketing Analytics	L
Business Architecture	marketing planning	L
Business Architecture	Measuring Value	H
Business Architecture	Product Lifecycle Management	H
Business Architecture	product planning	L
Business Architecture	Risk Profiling	L
Business Architecture	strategic planning	H
Business Architecture	Strategic Roadmapping	H
Business Architecture	Transition planning	M
General	Ethics, Integrity, Discretion, Confidentiality	H



General	Leadership communications (writing & speaking)	H
General	Negotiation & compromise	H
Implementation Architecture	Asset Management	L
Implementation Architecture	portfolio management	M
Implementation Architecture	Project Portfolio Management	L
Implementation Architecture	Release Management	L
Implementation Architecture	Requirements Engineering	H
Implementation Architecture	Test Planning & Defect Management	L
Information Architecture	Business Analytics	H
Information Architecture	Data Aggregation & Mining	M
Information Architecture	Data Architecture	H
Information Architecture	Data Mapping, Translation, and Integration	L
Information Architecture	Information Modeling	M
Information Architecture	Metadata Modeling	H
Operational Architecture	Domain (Healthcare - Knowledge Driven Model)	L
Operational Architecture	Domain (Manufacturing - Supply Chain Model)	L
Operational Architecture	Domain (Service Industry Model)	L
Operational Architecture	Executing Strategy	H
Operational Architecture	Functional Model (Back Office Processes)	M
Operational Architecture	Functional Model (ERP Processes)	M
Operational Architecture	Introduction to Design Patterns	H
Operational Architecture	Lean & Six Sigma Techniques	L
Operational Architecture	Metrics Modeling (KPIs through Modeling Performance)	H
Operational Architecture	Process Modeling	H
Technical Architecture	Hardware Platforming (Servers, Blades, Grid Computing)	M
Technical Architecture	Network Planning (basic knowledge of OSI stack)	L
Technical Architecture	OS Models (Unix, Windows, Ubuntu, Linux)	M
Technical Architecture	Platforming integration	M
Technical Architecture	System Development Lifecycle Mgmt	H
Technical Architecture	technology planning	M
Technical Architecture	Utility Services (Mail, Messaging, Collaboration)	M
Technical Architecture	Utility Services (Security - perimeter, network, data)	H

What will be the jobs landscape for graduates in 2011 and beyond who will graduate from these programs in government, business, military and elsewhere?

Job Projections

Domain	Opportunity
Agriculture, Food and Natural Resources	H
Architecture and Construction	L
Arts, Audio-Video Technology and Communications	H
Business, Management and Administration	H
Education and Training	H
Finance	M
Government and Public Administration	H
Health Science	H
Agriculture, Food and Natural Resources	H
Hospitality and Tourism	H
Human Services	H
Information Technology	H
Law, Public Safety, Corrections and Security	M
Manufacturing	H
Marketing, Sales and Service	H
Science, Technology, Engineering and Mathematics	H
Transportation, Distribution and Logistics	H

General questions for all committees – what are possible future directions for the overall EA Advisory Group? Are there other directions we should consider beyond our four initial areas of development (undergraduate education, graduate education, professional development, and research)?

Possible Future Directions for the EA Advisory Group:

- Link the programs to the value-delivery model which is evolving under the stewardship of CAEAP
- Elevate the focus of EA beyond technology
- Shift the discussion about the business-IT divide to engage the business analysts and business architects to link requirements management to a customer focus discipline.
- Change the mindset around how technology should be introduced (infused if you will) into the business planning & operations. Promote transparency and simplicity instead of marshalling forward with indoctrination programs to gain adoption.
- Bring real world problems in to the EA curriculum to stretch the thinking of students and staff.
- Engage the EA staff & students in projects that promote collaboration over competition (true professionals share their knowledge and assist / mentor junior/associate members).



Should this meeting evolve into an annual conference? If so, should be partner with an existing conference or start a new one?

We believe the conference could evolve into an advisory board that oversees EA education best practices that provides a EA "stamp" of approval for coursework and possibly sponsor innovation and/or improvements to further the educational offerings.

Additional Thoughts on Alignment with CAEAP's Professional Practice Guide & EA Curriculum:

CAEAP Practice Guide (Areas of Focus)	Undergrad	Graduate	Continuing Ed
The Profession of Enterprise Architecture	X	X	X
Standard Enterprise Architect Competency Framework	X		
Accreditation Standards for EA Frameworks and Methodologies			X
Standards for Recognized Professional Qualifications	X		X
Getting established in an Enterprise Architecture Practice	X	X	X
Growing Your Practice			X
Standards for Accreditation in Education	X	X	
Professional Registry for Enterprise Architecture Professionals			X
Enterprise Architecture Branding and Marketing			X
Awareness and Recognition for EA Professionals		X	X
Written Examination Above and Beyond Certification Requirements	X	X	X
Legal Frameworks		X	X
The Role of Apprenticeship in Enterprise Architecture			X
EA Education and Certification	X	X	X
Enterprise Architecture and It's Relationship with Other Architecture Streams	X	X	X
Enterprise Alignment & Business/IT Alignment;	X	X	X